

EcoStruxure Value Advisor

Schneider Electric — Singapore Emissions-to-Dollar Calculator

Thales Solutions Asia Pte Ltd — Singapore HQ

Singapore Headquarters — Sustainability & Cost-Savings Illustration

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Executive summary

What matters	Figure
Current annual electricity bill	S\$1,033,884 (3,240,000 kWh/year, estimated)
Current Scope 1 + 2 emissions	1,302 tCO₂e (Scope 1 = 0, not modelled — see below)
Recommended illustrative investment	S\$500,000
Expected saving, Year 1	S\$176,432
Expected saving, 10 years (cumulative range)	S\$1,255,725 – S\$2,608,044
Payback period	2.8 years
Recommended first step	LED lighting retrofit — 0.3 yr payback, S\$3,102/month

Confidence: Low (public-data estimate, not a metered bill) — see “Your Scope 1, 2 & 3 estimates used” and “Limitations” below for exactly what that means and why.

About this report

Prepared for Thales from public information about your Singapore headquarters, run through the same EcoStruxure Value Advisor calculation engine a Schneider advisor would use with your own metered data. It exists to show what the tool can do for your organisation specifically — not to claim knowledge of your actual energy bills, which Schneider does not have access to.

Two categories of number appear below:

- Company facts (address, sector, floor area, building certification) — each cited to a public source.
- Calculator outputs (emissions, cost, savings, payback) — computed by the tool from those facts, using Singapore-specific data (EMA grid factor, SP Group tariff, NEA carbon tax schedule) already built into the calculator.

Because no metered electricity bill is publicly available, the calculator estimated your electricity use from floor area × your sector's average energy intensity — its standard fallback for exactly this situation. That's why every figure below carries a “Low” confidence rating: it's a rough, defensible illustration, not a substitute for your own data. Confidence sharpens automatically once real kWh figures are entered.

Your Scope 1, 2 & 3 estimates used

Before any results below: this is exactly what was assumed to produce them, in plain terms.

Category	What was assumed	Basis
Scope 2 — Electricity use	Estimated 3,240,000 kWh/year = floor area (18,000 m²) × Manufacturing sector-average intensity (180 kWh/m²/yr)	Calculator's built-in floor-area fallback — no metered electricity bill is publicly available for this site
Scope 2 — Grid emission factor	0.402 kg CO ₂ /kWh	EMA Singapore, Grid Emission Factor 2024
Scope 1 — Fuel, vehicle fleet, backup generators	Assumed zero — not modelled	No public data on Thales's onsite fuel use, company vehicles, or backup generators at this site; left blank rather than guessed
Scope 1 — Refrigerants (fugitive emissions)	Assumed zero — not modelled	No public data on refrigerant type or annual top-up volume for this site's ACMV systems; left blank rather than guessed
Scope 3 — Commuting headcount	1,200 employees (same estimated headcount as Company snapshot)	Estimated split of Thales's 2,000+ Singapore-wide headcount across 3 sites — see Company snapshot
Scope 3 — Commuting distance & mode	15 km one-way (Singapore average) × 2 (round trip) × 250 working days/year × 0.107 kg CO ₂ e/km — the calculator's default blended rate (50% public transport at 0.044 kg/km, 50% private car at 0.17 kg/km), not a Thales-specific mode split	Calculator default (data/scope3_factors.json): distance/working-days per PRD spec; car factor DEFRA-matched, public-transport factor SMRT/LTA-derived (blended MRT+bus per 2024 ridership share)
Scope 3 — Result	1,200 × 30 km × 250 days × 0.107 kg/km = 963.0 tCO ₂ e/year	Calculated by the formula above — not a separate assumption

Company snapshot

Field	Value	Source
Registered address	12 Ayer Rajah Crescent, Thales Building, Singapore 139941	Thales Group / SGPBusiness company records
Sector classification	Manufacturing	Site houses Thales's Manufacturing Competence Centre (banking cards & passport datapages) plus avionics MRO —

Field	Value	Source
		classified Manufacturing rather than pure office, per Thales Group's own description of the site
Singapore headcount	2,000+ across 3 Singapore locations (this HQ site estimated at ~1,200)	Thales Group Singapore country page; single-site split is an estimate, not separately disclosed
Floor area used in this illustration	18,000 m ² (estimated)	Not publicly disclosed — estimated from a combined manufacturing + office/R&D site of this scale; refine with your own floor plan for an accurate result
Building certification	Not confirmed in public search results	n/a — no assumption made either way

Your current footprint & cost, as modelled

Metric	Value
Scope 1 (direct — fuel, fleet, refrigerants)	0 tCO ₂ e — not modelled: no public data on this site's fuel, vehicles, generators, or refrigerants (see "Your Scope 1, 2 & 3 estimates used" above). This is not a claim of zero real emissions.
Scope 2 (electricity, location-based)	1,302 tCO ₂ e
Total Scope 1 + 2	1,302 tCO ₂ e
Scope 3: employee commuting (estimated) — context only, not monetised	963 tCO ₂ e
Current annual electricity cost	S\$1,033,884
Energy intensity	180 kWh/m ² /year (sector: best 80, average 180, poor 320)

Your modelled energy intensity lands at the sector average (180 kWh/m²/year) — a direct mathematical consequence of estimating usage from the sector average itself, not a claim about your real efficiency. Your actual position could be meaningfully better or worse once real data replaces this estimate. No renewable electricity coverage (on-site solar, REC, or PPA) was entered, so location-based and market-based Scope 2 are identical here.

What you're already saving

No specific Energy Conservation Measures were marked as "already implemented" in this illustration — that information isn't publicly available, so nothing has been assumed or credited.

What you could potentially save

The 17% figure below is not a black box — it's the combined, diminishing-returns-adjusted effect of these 8 specific catalog measures (full detail, cost, and payback for each is in the "Full Energy Conservation Measure breakdown" table right after this one):

Waste heat recovery on process equipment; Variable-speed drives on HVAC fans & pumps; BMS-based scheduling & setpoint optimization; Compressed air leak survey & repair; Variable-speed drives on process motors, pumps & fans; LED lighting retrofit; Occupancy sensors & daylight harvesting controls; Sub-metering & power-quality monitoring.

Metric	Value
Bottom-up energy saving rate	17% — see the 8 measures listed above and detailed below
Monthly \$ saved (range)	S\$9,557 – S\$19,849
Year-1 \$ saved (mid-point)	S\$176,432
10-year cumulative financial benefit (range)	S\$1,255,725 – S\$2,608,044
Illustrative payback period	2.8 years, against an illustrative S\$500,000 investment (see Limitations)
Rough investment range implied by these 8 measures	S\$145,000 – S\$560,000

Top 3 recommended measures

Ranked by payback (cost ÷ annual saving) — least effort, most gain first.

Measure	Monthly \$ saved	Payback	Cost range
LED lighting retrofit	S\$3,102	0.3 yr	S\$5,000–S\$20,000
Occupancy sensors & daylight harvesting controls	S\$1,379	0.8 yr	S\$5,000–S\$20,000
BMS-based scheduling & setpoint optimization	S\$1,292	0.8 yr	S\$5,000–S\$20,000

Full Energy Conservation Measure breakdown

Every catalog measure relevant to your sector that isn't marked implemented — 8 in total.

Measure	Building area	\$/year saved	Payback
Waste heat recovery on process equipment	Process heating / equipment	S\$52,728	3.6 yr
Variable-speed drives on HVAC fans & pumps	HVAC / space cooling	S\$31,017	1.6 yr
BMS-based scheduling & setpoint optimization	HVAC / space cooling	S\$15,508	0.8 yr
Compressed air leak survey & repair	Compressed air & motors	S\$9,305	1.3 yr
Variable-speed drives on process motors, pumps & fans	Compressed air & motors	S\$24,813	2.0 yr
LED lighting retrofit	Lighting	S\$37,220	0.3 yr

Measure	Building area	\$/year saved	Payback
Occupancy sensors & daylight harvesting controls	Lighting	S\$16,542	0.8 yr
Sub-metering & power-quality monitoring	Other / miscellaneous	S\$3,877	3.2 yr

Note — these rows won't sum to the headline figure, on purpose: Adding up the “\$/year saved” column gives S\$191,010/year — about 8% higher than the Year-1 mid-point figure stated earlier. That's expected, not an error: each row is that measure's saving computed on its own, while the headline % and \$ figures use the calculator's proper combination across measures that share the same building area (e.g. several HVAC measures together don't each deliver their full saving independently — diminishing returns apply). Use this table to compare and prioritise individual measures, not to be added into a combined total.

Benchmark position

Sector: Manufacturing. Best-in-class 80 kWh/m²/yr — sector average 180 — poor 320. *(Source: BCA Building Energy Benchmarking Report, go.gov.sg/bca-bebr-data.)*

Proof this is achievable — a real Schneider deployment, for reference

Samwoh Smart Hub, Singapore: A Schneider Electric-instrumented industrial facility saving ~S\$220k/year (~4-year payback) and running 84% below its 2005 baseline — Singapore's first BCA Green Mark Platinum Positive Energy industrial building (2024).

Not necessarily at your exact position on the curve above — shown to demonstrate real, achievable results, not as a direct comparison point.

Regulatory exposure check

Not a direct carbon taxpayer in this illustration. Singapore's carbon tax (Carbon Pricing Act) applies only to facilities with ≥25,000 tCO₂e/year of direct (Scope 1) emissions — about 50 large industrial sites nationally. Your only exposure is a small pass-through already reflected in the electricity tariff used above, so no separate carbon-tax saving line is shown.

Assumptions & data sources

This table is the calculator's own raw assumptions output, unedited.

Assumption	Value	Source
Grid Emission Factor	0.402 kg CO ₂ /kWh	EMA Singapore, Grid Emission Factor 2024
Electricity Tariff	S\$0.3191/kWh	SP Group 2026-Q3
Diesel Emission Factor	2.68 kg CO ₂ e/litre	Diesel/petrol: Singapore Emission Factors Registry (SEFR) / NEA. CNG: UK DEFRA 2023 conversion factors via Clima ^{tiq} (2562.57 kg CO ₂ e/tonne stationary combustion), verified 2026-09-17 —

Assumption	Value	Source
		was previously 2.15, understated by ~19% relative to this sourced figure and to the stoichiometric pure-methane combustion value (~2.75 kg CO ₂ /kg CH ₄); CNG is ~85-95% methane so a value just below 2.75 is expected.
Carbon tax liability	Not a direct taxpayer (pass-through only)	NEA Carbon Pricing Act — applies to direct (Scope 1) emitters only, ~50 facilities nationally
Current Rate (2024-2025)	S\$25 per tCO ₂ e	NEA / MSE Singapore
Legislated Rate (2026-2027)	S\$45 per tCO ₂ e	NEA / MSE Singapore
Indicative Rate (2030)	S\$65 per tCO ₂ e (base)	NEA / MSE Singapore — indicative, not yet legislated
Energy Saving Applied	17% (bottom-up from your sector's remaining Energy Conservation Measures)	Energy Conservation Measures catalog, net of measures already implemented — see data/ecm_catalog.json
EEG Grant	Up to S\$30,000 (70% co-fund) — informational only, NOT included in the \$ savings/payback above (case-by-case eligibility)	EnterpriseSG, valid to 2028-03-31
Scope 3 Method	Spend-based (EEIO) — quantified only, not included in \$ saving	DEFRA 2026 GHG Conversion Factors (business travel updated); DEFRA 2024 (freight — see freightNote); US EPA EEIO v2.0 (spend-based fallback)
Energy End-Use Breakdown	Sector-typical split shown	International/market-survey composite (no Singapore-specific end-use metering study was found) — see per-sector citations below. Not a regulator-published figure like the grid emission factor or carbon tax schedule; shown to the user as an indicative default, editable.

"Energy saving applied" above is the same 17% from the same 8 measures named earlier in "What you could potentially save" and detailed in "Full Energy Conservation Measure breakdown."

Summary

Thales Solutions Asia Pte Ltd (Singapore HQ) — a Manufacturing business with an estimated 18,000m² across Singapore — emits approximately 1,302 tCO₂e/year across Scope 1 and 2. Implementing the measures identified below, this illustration estimates a saving of approximately S\$176,432 per year on energy costs. Singapore's carbon tax doesn't apply directly to this facility at this scale — its cost is already embedded in the electricity tariff used above. Over 10 years, the cumulative financial benefit is estimated at S\$1,255,725 to S\$2,608,044, against an illustrative S\$500,000 investment, for a payback of approximately 2.8 years. The Scope 3 footprint of ~963 tCO₂e/year (largely employee commuting,

estimated from headcount) is identified but not included in the financial saving — tracking and reporting it is useful for SBTi and sustainability-reporting purposes.

Limitations of this illustration

In the interest of being fully transparent about what this document is and isn't:

- **The ECM breakdown table doesn't sum to the headline figure, on purpose.** Adding up the "\$/year saved" column gives S\$191,010 — about 8% above the Year-1 saving stated earlier. Each row is that measure's saving computed independently; the headline figure properly discounts for measures that share the same building area (e.g. several HVAC measures) so it doesn't double-count overlapping savings. This is a structural property of how the calculator computes and displays these two figures, not specific to this report.
- **Scope 1 is shown as 0 tCO₂e** because no public data exists on this site's fuel use, company vehicles, backup generators, or refrigerants — it was left unmodelled rather than guessed. This is *not* a claim that your real direct emissions are zero; a real HQ of this size almost certainly has some (backup generator testing, refrigerant top-ups, etc.).
- **Confidence is rated Low** for every figure in this report, which in the calculator's own terms means a $\pm 35\%$ range around every \$ and tCO₂e figure shown — reflecting that electricity use was estimated, not metered. This narrows automatically once real kWh data is entered.
- **Cost figures are generic order-of-magnitude tiers** (e.g. S\$5,000–20,000 for a "low" cost measure), applied uniformly across the whole measure catalog — not a site-specific quote for your building.
- **The illustrative investment figure** (S\$500,000) used to compute the payback period above was chosen by us as a round placeholder within the range the selected measures imply — it is not sourced from, or confirmed by, your organisation.
- **This uses the calculator's single-site methodology** (its SME mode), chosen because this illustration covers one HQ site. The calculator also has a multi-site portfolio mode, which may produce a different picture if your Singapore operations span multiple sites or buildings.
- **Sector benchmark bands** (best-in-class / average / poor kWh/m²/yr) come from the BCA Building Energy Benchmarking Report (go.gov.sg/bca-bebr-data) for this sector; see "Benchmark position" above.
- **This illustration has no knowledge of measures you may have already implemented** beyond what's publicly disclosed above. If you've already retrofitted lighting, upgraded HVAC controls, or made other efficiency investments, those should be marked as implemented in the live tool — doing so would reduce the "potential savings" figures above (they'd move to "already saving" instead) and is a more accurate picture than this document alone.
- This is a calculator demonstration, not a professional energy audit. A Schneider advisor, working from your real utility bills and a site walkthrough, would produce a materially more precise version of every figure in this report.

Prepared by EcoStruxure Value Advisor for Schneider Electric. This illustration uses public information and the calculator's built-in Singapore reference data; it is not a quotation and does not represent knowledge of your actual metered energy use. A Schneider advisor can refine every figure above with your real utility bills, site audit findings, and confirmed implemented measures.